The Mpemba Effect as a Finite-Response Coherence Transition

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The Mpemba effect, in which hot water freezes faster than cold water under certain conditions, remains an unresolved problem in non-equilibrium thermodynamics. We propose a unifying interpretation based on finite-response coherence dynamics. In this framework, freezing is not determined solely by temperature but by the ability of the system to transition into a coherent phase under finite response constraints. A dimensionless coherence parameter is introduced, governed by the ratio of intrinsic response time to external driving timescale. This approach naturally unifies previously proposed mechanisms, including evaporation, convection, supercooling, and degassing, as contributions to a single response-controlled transition criterion. Testable predictions are provided.

INTRODUCTION

The Mpemba effect refers to the counterintuitive observation that, under certain conditions, initially hotter water can freeze faster than initially colder water. Despite numerous proposed explanations, including evaporation, convection, and dissolved gas effects, no consensus mechanism has emerged.

A key limitation of existing approaches is the assumption that freezing is determined primarily by temperature evolution. However, freezing is inherently a phase transition requiring the formation of coherent structure, suggesting that dynamical response properties may play a central role.

FINITE-RESPONSE COHERENCE FRAMEWORK

We introduce a dimensionless coherence parameter

$$\chi = \frac{1}{1 + \left(\frac{\tau_{\mathrm{resp}}}{\tau_{\mathrm{drive}}}\right)^{p}}, \tag{1}$$

where τ_{resp} characterizes the intrinsic structural response time of the system, τ_{drive} is the effective cooling or forcing timescale, and $p > 0$.

The transition to a frozen phase occurs when the system becomes sufficiently coherent,

$$\tau_{\mathrm{resp}} \lesssim \tau_{\mathrm{drive}} \quad \Longleftrightarrow \quad \chi \sim \mathcal{O}(1). \tag{2}$$

This condition generalizes the freezing criterion beyond temperature alone.

INTERPRETATION OF THE MPEMBA EFFECT

Within this framework, initially hot water can freeze faster than cold water due to differences in response dynamics:

- Heating reduces structural memory and dissolved gases, lowering τ_{resp}.
- Cold water may remain in metastable configurations, increasing τ_{resp}.
- Supercooling corresponds to $\chi \ll 1$ despite low temperature.

Thus, the system that reaches $\chi \sim 1$ first freezes first, regardless of initial temperature.

UNIFICATION OF PROPOSED MECHANISMS

Previously proposed explanations map naturally onto the response framework:

- Evaporation modifies τ_{drive}.

- Convection enhances effective response.
- Degassing reduces τ_{resp}.
- Container and boundary effects alter coherence pathways.

These are not independent mechanisms but contributions to a single governing ratio $\tau_{\mathrm{resp}}/\tau_{\mathrm{drive}}$.

EXPERIMENTAL PREDICTIONS

The framework yields clear predictions:

1. Pre-heating reduces τ_{resp} and increases freezing rate.
2. Mechanical agitation accelerates freezing by enhancing response.
3. Equalizing structural conditions removes the Mpemba effect.
4. Freezing time is not a single-valued function of temperature.

CONCLUSION

The Mpemba effect is reinterpreted as a finite-response coherence transition rather than a purely thermodynamic anomaly. This provides a unified, physically grounded explanation and suggests new experimental tests of non-equilibrium phase transitions.